

Electricity — MCQ Worksheet

Science • Chapter 11 • 25 Questions, 1 mark each — Total: 25 marks

Instructions: Choose the correct option (A / B / C / D) for each question. Each question carries 1 mark. There is no negative marking. Suggested time: 30 minutes.

Q1. The SI unit of electric current is:

- (A) Volt
- (B) Ampere
- (C) Coulomb
- (D) Ohm

Q2. 1 ampere = 1 coulomb / ____

- (A) Hour
- (B) Minute
- (C) Second
- (D) Day

Q3. Conventional current flows from:

- (A) Negative to positive terminal outside cell
- (B) Positive to negative terminal outside cell
- (C) No fixed direction
- (D) Only inside the cell

Q4. The unit of electric potential difference is:

- (A) Ohm
- (B) Volt
- (C) Watt
- (D) Joule

Q5. An ammeter is connected in:

- (A) Parallel
- (B) Series
- (C) Either
- (D) None

Q6. A voltmeter is connected in:

- (A) Series
- (B) Parallel
- (C) Both
- (D) Open circuit only

Q7. Ohm's law states that V is proportional to I provided:

- (A) Resistance is zero
- (B) Temperature is constant
- (C) Current is zero
- (D) Voltage is zero

Q8. The SI unit of resistance is:

- (A) Ampere
- (B) Volt
- (C) Ohm
- (D) Coulomb

Q9. The resistance of a wire depends on:

- (A) Length only
- (B) Area only
- (C) Material only
- (D) Length, area, material and temperature

Q10. Resistance R of a wire is given by $R = \rho L/A$. Here ρ is the:

- (A) Volume
- (B) Density
- (C) Resistivity
- (D) Conductivity

Q11. In series combination of resistors, the equivalent resistance is:

- (A) $R_1 + R_2 + R_3$
- (B) $1/R_1 + 1/R_2 + 1/R_3$
- (C) $R_1 \times R_2 \times R_3$
- (D) Always less than R_1

Q12. In parallel combination of resistors, the equivalent resistance:

- (A) Equals $R_1 + R_2$
- (B) Is less than the smallest resistor
- (C) Is greater than the largest
- (D) Is zero

Q13. Domestic appliances are connected in parallel because:

- (A) Each gets full voltage and works independently
- (B) It saves wires
- (C) Series wiring is dangerous
- (D) Parallel uses less power

Q14. Electrical energy dissipated in a resistor is given by:

- (A) $H = IR^2t$
- (B) $H = I^2Rt$
- (C) $H = IRt$
- (D) $H = I^2R/t$

Q15. Power is given by $P = \underline{\hspace{1cm}}$?

- (A) VI
- (C) I/V

- (B) V/I
- (D) $V + I$

Q16. 1 kilowatt-hour (kWh) equals:

- (A) $3.6 \times 10^3 \text{ J}$
- (C) $3.6 \times 10^6 \text{ J}$

- (B) $3.6 \times 10^5 \text{ J}$
- (D) $3.6 \times 10^9 \text{ J}$

Q17. A 100 W bulb at 220 V draws a current of approximately:

- (A) 0.45 A
- (C) 4.5 A

- (B) 2 A
- (D) 10 A

Q18. The function of an electric fuse is to:

- (A) Increase voltage
- (C) Break the circuit on excess current

- (B) Reduce voltage
- (D) Store charge

Q19. Filament of an electric bulb is made of:

- (A) Copper
- (C) Iron

- (B) Tungsten
- (D) Aluminium

Q20. The most common reason a fuse melts is:

- (A) Overcurrent due to short circuit or overloading
- (C) No current

- (B) Low voltage
- (D) High resistance always

Q21. An earth wire in a 3-pin plug is connected to:

- (A) The electric appliance casing/body
- (C) The live wire

- (B) The neutral wire
- (D) Nothing

Q22. In a household, the live wire is colour-coded:

- (A) Red
- (C) Green

- (B) Black
- (D) Yellow

Q23. Two resistors of 4Ω and 6Ω in series have equivalent resistance:

- (A) 2.4Ω
- (C) 24Ω

- (B) 10Ω
- (D) 2Ω

Q24. Two resistors of 4Ω and 6Ω in parallel have equivalent resistance:

- (A) 2.4Ω
- (C) 24Ω

- (B) 10Ω
- (D) 12Ω

Q25. Power dissipated in a resistor of 5Ω carrying 2 A is:

- (A) 10 W
- (C) 2.5 W

- (B) 20 W
- (D) 100 W